Photonic resistive switching: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

With $\beta = (2\pi/\lambda_0)(n + i\kappa)$, the field magnitude acquires $e^{-2\pi\kappa L/\lambda_0}$. Squaring gives $T = e^{-4\pi\kappa L/\lambda_0}$. At $\kappa = 0.01$ the exponent is -0.1621, so $T_H \approx 0.8503$; at 0.08 it is -1.2972, so $T_L \approx 0.2733$. Thus $ER \approx 10\log_{10}(0.8503/0.2733) = 4.929\,\mathrm{dB}$ and $IL_H \approx 0.704\,\mathrm{dB}$. The effective mode includes field distribution in multiple materials and boundaries; a bulk index alone does not determine it.

E02

For constant V and I, $E = VI\Delta t = (1.2)(10\times10^{-6})(20\times10^{-9}) = 2.4\times10^{-13}\,\mathrm{J} = 240\,\mathrm{fJ}$. Driver and interconnect loss, optical probe energy, reset and sense circuitry are excluded. A changing waveform requires integrating the measured product instead.

E03

$G = 0.3(77.5\,\mu\mathrm{S}) = 23.25\,\mu\mathrm{S}$, so $R = 1/G \approx 43.0\,\mathrm{k\Omega}$. The pair $T_1 = 0.1, T_2 = 0.2$ gives the same sum and hence the same conductance in the model. A conductance value or step alone does not distinguish these channels or determine a unique atomic arrangement.

E04

The probability of zero failures is $(1-p)^{100}$. Solving $(1-p)^{100} = 0.05$ yields the one-sided 95% upper bound $p = 1 - 0.05^{1/100} \approx 0.0295$, or 2.95%. This calculation assumes independent observations with a common p, and applies to the stated test only. Correlated failures across cycles or devices violate the independence assumption; 100 correlated outcomes cannot simply be inserted in this formula. The result does not prove zero failure or retention at untested times.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

Photonic resistive switching: Worked solutions (continued)

E05

Optical transmission depends on mode overlap, absorption, scattering and resonance. A conductive change may add absorption in one waveguide but shift or enable a transmitting resonance in a different MIM structure. Control at least wavelength, input polarization, input power, coupling geometry, temperature and the electrical state preparation before comparing. LRS refers to electrical resistance, not a universal optical ordering.

E06

The absolute difference is $0.25 - 0.20 = 0.05$, i.e. five percentage points. The ratio is $0.25/0.20 = 1.25$, and $10\log_{10}(1.25) \approx 0.969\,\mathrm{dB}$. Without the baseline R_1, the same difference could give a different ratio, so dB cannot be inferred from the difference alone.

E07

A hybrid guide uses a low-index gap between a high-index guiding region and metal; MIM uses an insulating gap between metal boundaries. Both can place optical field near a small active region, but their field distributions and metal overlap depend on their geometry and cannot be ranked without a specified design. Greater field in lossy metal can increase interaction but also dissipate more power. The simple planar-interface TM surface-mode dispersion assumes homogeneous infinite half-spaces and does not describe every finite, layered or patterned guide.

E08

Apply a specified electrical write pulse; record a low-disturbance optical read; set device bias to zero and wait a stated time at controlled temperature; repeat the read and compare with a control and the initial state. Repeat across cycles, devices and wait times while monitoring probe-induced changes. A sweep hysteresis is measured while voltage varies and may include a transient response; it does not establish state survival at zero bias.

Photonic resistive switching: Worked solutions (continued)

E09

From E01, $T_H \approx 0.8503$. Under the stated independent fixed-coupling model, $T_{\text{total}} = T_H^3 \approx 0.615$ and $IL_{\text{total}} = 3(0.704) \approx 2.112\,\text{dB}$. Extra coupler loss, reflections or coherent interference make the simple product insufficient.

E10

$P_{\text{avg}}/f = 12.5 \times 10^{-9}/10^6 = 12.5 \times 10^{-15}\,\text{J} = 12.5\,\text{fJ}$ per operation on average under the stated complete-period measurement and average one-operation-per-period rate. Instantaneous power can vary: its time average already includes active and idle intervals. Peak or active-only power omits part of that averaging and cannot simply replace P_{avg}. This system average need not equal an individual write pulse's device energy. Measure synchronized device voltage and current and integrate $\int V I\, dt$ over the pulse, separately reporting driver and peripheral energy.

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